

CHM 151: CHEMISTRY AROUND US



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Chemistry Around Us

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1: Creating Accounts

Creating Libretext ADAPT Account

You will use the ADAPT system of LibreText to submit your assignments and you will use the Single Sign On option through your school (@vikes.csuohio.edu) email. **Do not create a private account within the ADAPT system.**

1. Go to <https://adapt.libretexts.org>. (Figure 1.3) Click **Log IN** (DO NOT REGISTER!)

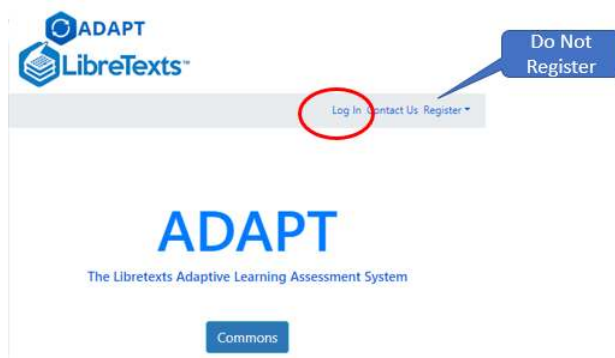


Figure 1.3: Click Log In. Do not Register.

2. Click Campus Login (**DO NOT FILL OUT FORM**) (figure 1.4).

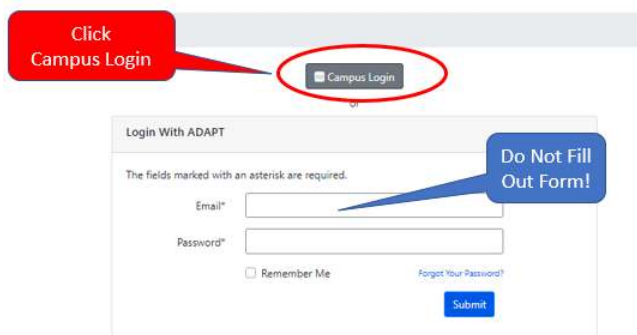


Figure 1.4: Click "Campus Login", this will allow you to set up a Single Sign On (SSO) with your UALR email account.

3. Click "Microsoft Active Directory" (figure 1.5).

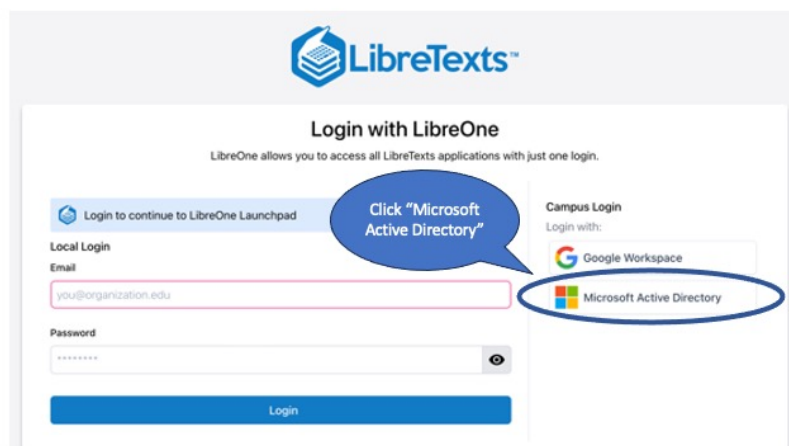


Figure 1.5: You will be able to log in with your CSU email through the Microsoft Active Directory option)

4. Choose your CSU email account and fill out form, choosing the student option (figure 1.6).

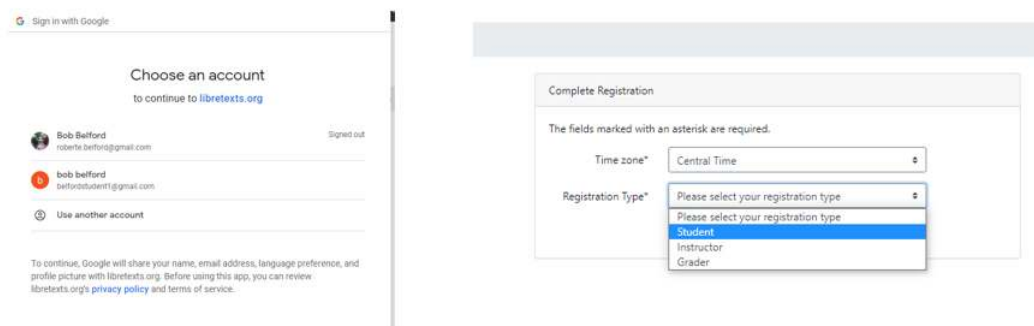


Figure 1.6: Be sure to set up the account with your UALR email address.

5. You now have an account and can enroll in a course.

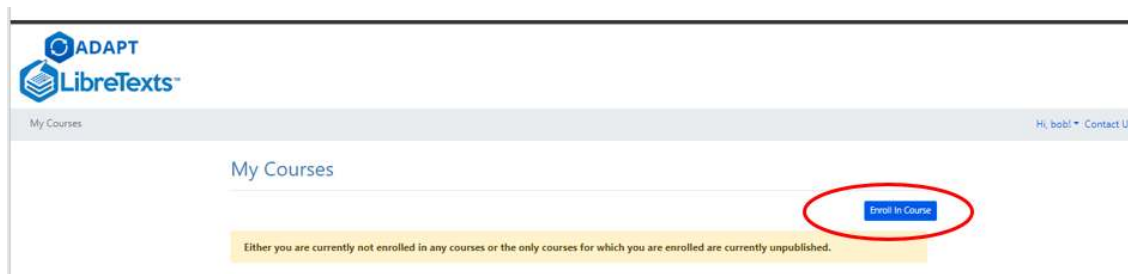


Figure 1.7: You now have an account within the ADAPT system and need to get an access code from your instructor to enroll in a course (Copyright; author via source)

6. You will need an access code for each course you take. This is provided by your instructor and can be found in Blackboard.

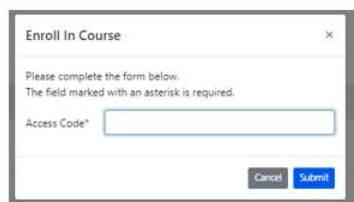


Figure 1.8: Insert your access code and you will become registered for a course.

7. You now can navigate to your course assignments

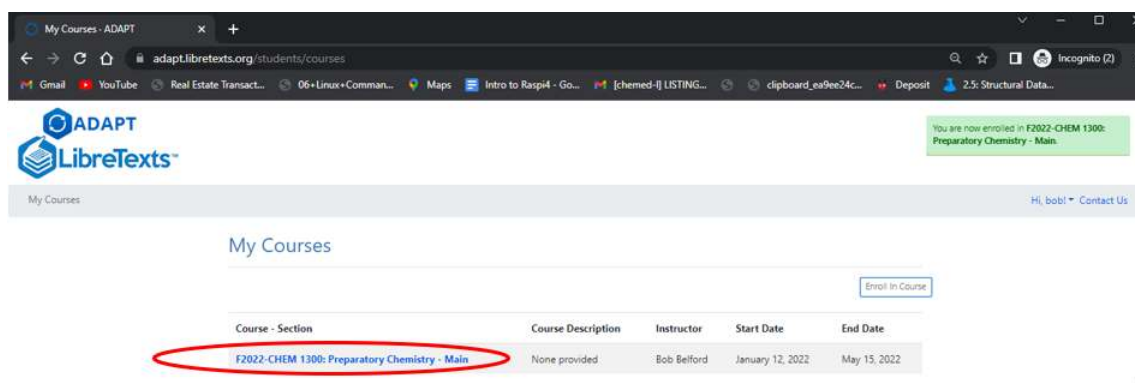


Figure 1.9: Copy and Paste Caption here. (Copyright; author via source)

Setting up Notifications

ADAPT can send an alert to your email when an assignment is due. This is advised as you will lost points for late assignments.

1. Next, Click on your name in the top right corner and go to Settings.

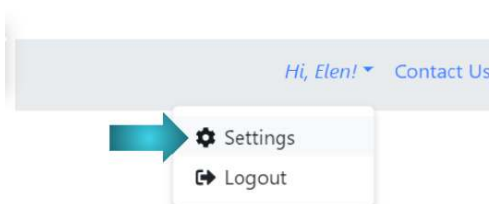


Figure 1.10: Open settings

2. Go to Notifications and choose the time option for your reminders. Click "Update"

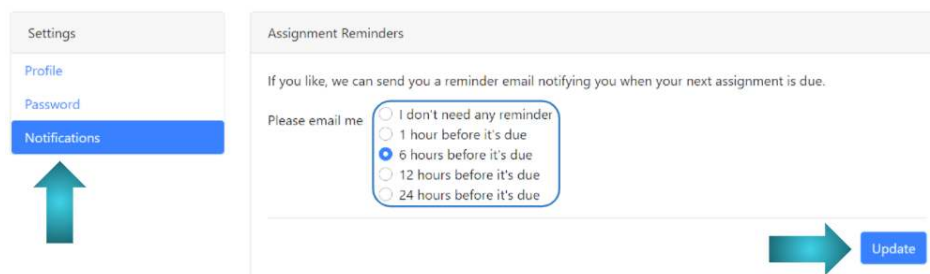


Figure 1.11: Copy and Paste Caption here. (Copyright; author via source)

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Licensing

A detailed breakdown of this resource's licensing can be found in [Back Matter/Detailed Licensing](#).

CHAPTER OVERVIEW

1: Properties and Measurement of Matter

1.1: Classification and Properties of Matter

1.2: All About the Elements

1.3: Measuring Matter

1.4: Unit Conversions

1.5: Density

1.6: Temperature

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1.1: Classification and Properties of Matter

This page is a draft and is under active development.

Learning Objectives

- Use physical and chemical properties to describe matter.
- Distinguish between physical and chemical changes.
- Describe the properties of each state of matter.
- Identify a sample of matter as an element, a compound, or a mixture (homogeneous/heterogeneous).
- Distinguish homogenous and heterogenous mixtures.

1.1.1 Physical and Chemical Properties

The properties that chemists use to describe matter fall into two general categories. Physical properties are characteristics that **describe** matter. They include characteristics such as size, shape, color, and mass. Many of these properties can be quantitative in nature. For example, quantitative physical properties of water would be the boiling point (100 °C / 212 °F) and melting point (0°C / 32 °F).



Figure 1.1.1: Physical Properties. (Copyright;<https://www.slideshare.net/cfoltz/ph...cal-properties>)

Chemical properties are characteristics that describe how matter **changes** its chemical structure or composition. An example of a chemical property is flammability—a material's ability to burn—because burning (also known as combustion) changes the chemical composition of a material. Oxidation, rusting, decomposition, and inertness are chemical properties as well. Click on this [video](#) and record the physical and chemical properties of the element sodium.



Figure 1.1.2: Port Royal, South Carolina oxidation of iron. Image courtesy of Elizabeth R. Gordon.

1.1.2 Elements and Compounds

Any sample of matter that has the same physical and chemical properties throughout the sample is called a substance. There are two types of substances. A substance that cannot be broken down into chemically simpler components is an **element**. Aluminum,

which is used in soda cans, is an element. A substance that consists of two or more atoms is called a **molecule**. Oxygen gas (O_2) is an example of a molecule because it contains two atoms of oxygen bonded together. A substance that can be broken down into chemically simpler components (because it has more than one element) is a **compound** (Figure 1.1.1). Water (H_2O) is a compound composed of the elements hydrogen and oxygen. Today, there are 118 elements in the known universe. In contrast, scientists have identified tens of millions of different compounds to date.

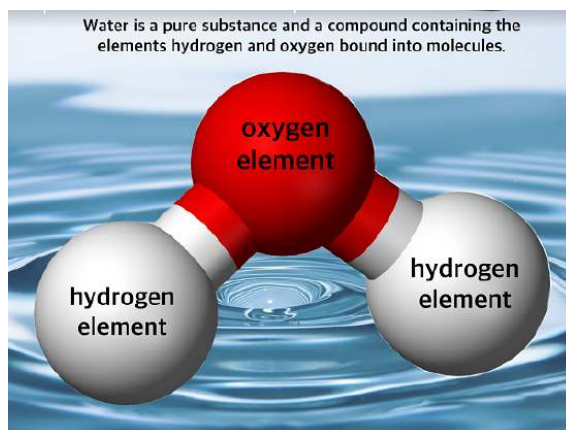


Figure 1.1.3: To a chemist, the pure substance water is a compound (it contains two kinds of atoms bound to one another in just one kind of molecule). An environmentalist might consider water "pure" even if it contains the normal amount of dissolved oxygen and carbon dioxide, but no other "pollutants". To a chemist, water containing oxygen is no longer a pure substance, but a mixture. In pure water, the ratio of hydrogen to oxygen atoms is always 2:1.

Sometimes the word *pure* is added to a substance, but this is not absolutely necessary. By definition, any single substance is pure.

The smallest part of an element that maintains the identity of that element is called an atom. Atoms are extremely tiny; to make a line 1 inch long, you would need 217 million iron atoms. The smallest part of a compound that maintains the identity of that compound is called a molecule. Molecules are composed of atoms that are attached together and behave as a unit. Scientists usually work with millions and millions of atoms and molecules at a time. When a scientist is working with large numbers of atoms or molecules at a time, the scientist is studying the macroscopic view of the universe.

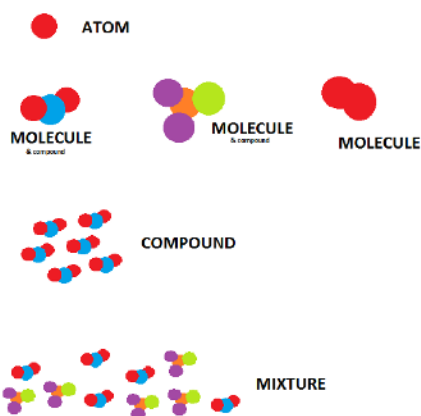


Figure 1.1.4: Appreciating elements, molecules and compounds.

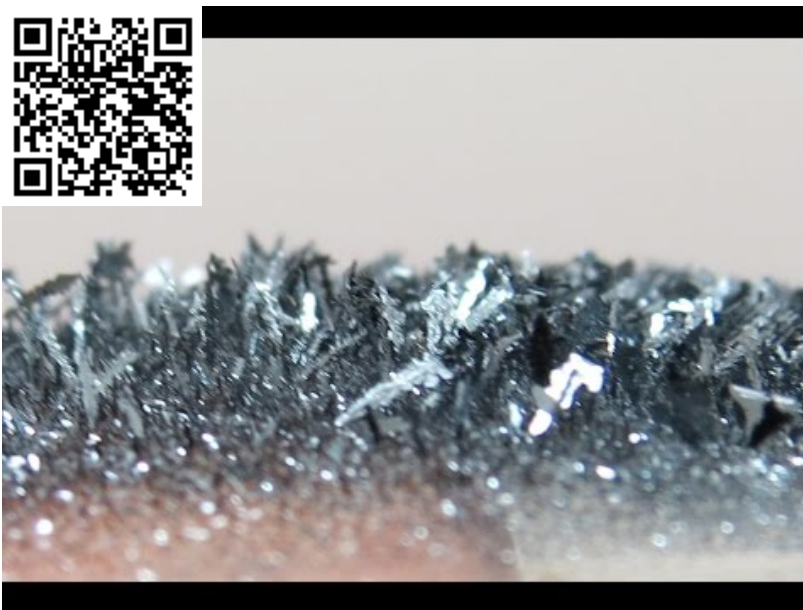
1.1.3 Mixtures

A material composed of two or more substances is a mixture. In a mixture, the individual substances maintain their chemical identities. Many mixtures are obvious combinations of two or more substances, such as a mixture of sand and water. Such mixtures are called **heterogeneous mixtures**. In some mixtures, the components are so intimately combined that they act like a single substance (even though they are not). Mixtures with a consistent composition throughout are called **homogeneous mixtures** (or

solutions). Sugar dissolved in water is an example of a solution. A metal alloy, such as steel, is an example of a solid solution. Air, a mixture of mainly nitrogen and oxygen, is a gaseous solution.

1.1.4 Phases

Another way to classify matter is to describe it as a solid, a liquid, or a gas, which was done in the examples of solutions. These three descriptions, each implying that the matter has certain physical properties, represent the three phases of matter. A solid has a definite shape and a definite volume. Liquids ordinarily have a definite volume but not a definite shape; they take the shape of their containers. Gases have neither a definite shape nor a definite volume, and they expand to fill their containers. We encounter matter in each phase every day; in fact, we regularly encounter water in all three phases: ice (solid), water (liquid), and steam (gas).



Video1.1.35: Sublimation and deposition of Iodine.

We know from our experience with water that substances can change from one phase to another if the conditions are right. Typically, varying the temperature of a substance (and, less commonly, the pressure exerted on it) can cause a phase change, a physical process in which a substance goes from one phase to another (Figure 1.1.3). Phase changes have particular names depending on what phases are involved, as summarized in Table 1.1.1.

Table 1.1.1: Phase Changes

Change	Name
solid to liquid	melting, fusion
solid to gas	sublimation
liquid to gas	boiling, evaporation
liquid to solid	solidification, freezing
gas to liquid	condensation
gas to solid	deposition

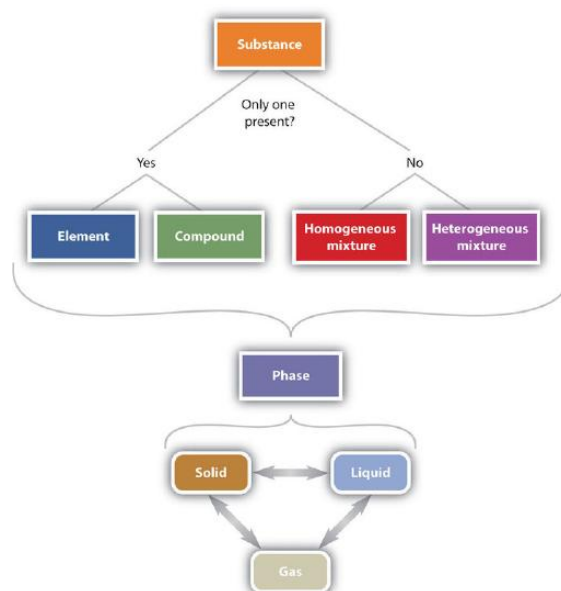


Figure 1.1.6: The Classification of Matter. Matter can be classified in a variety of ways, depending on its properties. (Cc BY-NC-SA; anonymous)

Contributors and Attributions

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